INCH-POUND

MIL-DTL-83522/1G
3 April 2023
SUPERSEDING
MIL-C-83522/1F
17 May 2006

MILITARY SPECIFICATION SHEET

CONNECTOR, PLUG, FIBER OPTIC, SINGLE TERMINUS, THREADED, (STEP-DOWN NOSE INTERFACE), LENSLESS, EPOXY, FOR 50/125 µm, 62.5/125 µm, AND 100/140 µm

Inactive for new design after 17 August 1999

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83522.

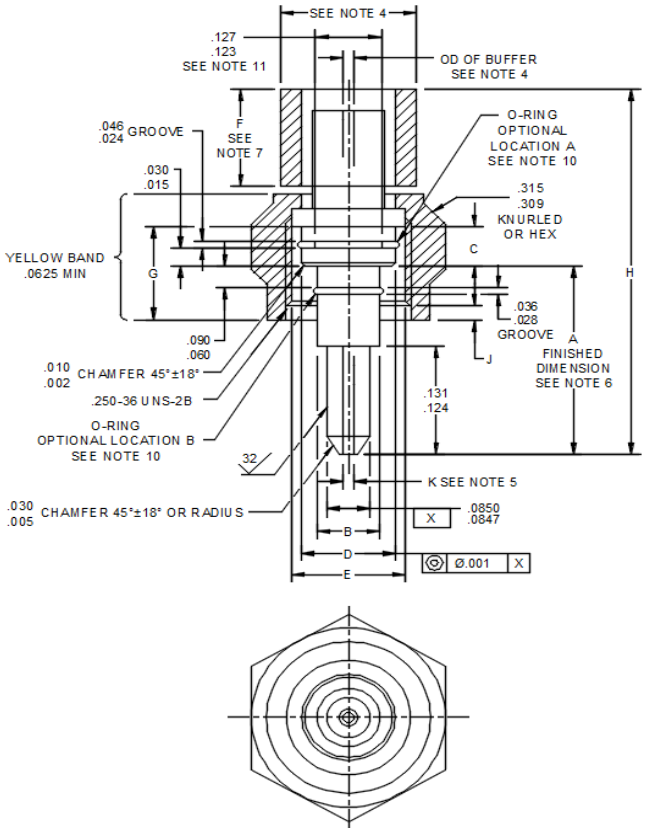


FIGURE 1. Dimensions and configuration (step-down nose).

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FSC 6060

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Inches	mm	Inches	mm
.001	0.02	.0625	1.588
.002	0.05	.0847	2.151
.005	0.13	.0850	2.159
.010	0.25	.090	2.29
.015	0.38	.123	3.12
.024	0.61	.124	3.15
.028	0.71	.127	3.23
.030	0.76	.131	3.33
.036	0.91	.250	6.35
.046	1.17	.309	7.85
.060	1.52	.315	8.00

K dimensions		
50/125 µm fiber	62.5/125 µm fiber	100/140 µm fiber
.1300 max .1275 min	.1300 max .1275 min	.1450 max .1425 min

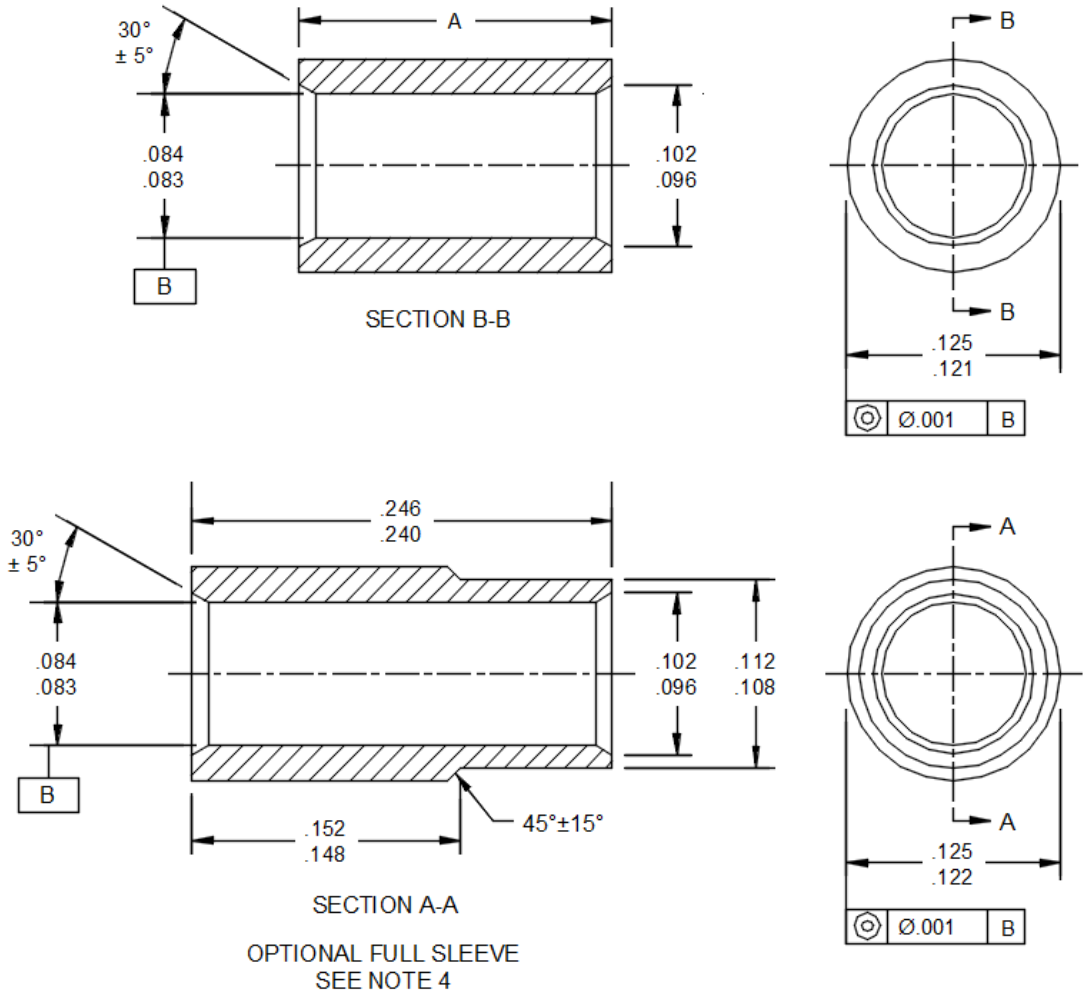
	Min	Max
A	.3858 (9.799)	.3863 (9.812)
C	.080 (2.03)	.100 (2.54)
D	.179 (4.55)	.182 (4.62)
E	.220 (5.59)	.233 (5.92)
F	.485 (12.32)	---
G	.183 (4.65)	.244 (5.69)
H	---	.150 (38.1)
J	.015 (0.38)	.045 (1.14)
B	Plug must not enter .118 hole gauge	Plug must enter .124 dia. hole gauge

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Dimensions for fiber optic cable shall be in accordance with MIL-PRF-85045.
5. K dimension is specified in micrometer. This dimension is critical to fiber alignment. Dimensions for fiber optic fiber shall be in accordance with MIL-PRF-49291.
6. Dimension A is obtained after product has been assembled.
7. Ferrule may be an integral part of component.
8. One full length sleeve to be used in conjunction with two step-down nose connectors.
9. One-half length sleeve to be used in conjunction with one step-down nose connector and one receptacle (active device).
10. O-ring is required. Component shall be machined for either O-ring location A or location B.
11. Dimension of O-ring area.

FIGURE 1. Dimensions and configuration - Continued.

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OPTIONAL FULL SLEEVE
SEE NOTE 4

	A Min	A Max
Full sleeve (see note 4)	.240 (6.10)	.246 (6.25)
One-half sleeve (see note 5)	.117 (2.97)	.124 (3.15)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. One full length acetal sleeve to be used in conjunction with two step-down nose connectors.
5. One-half length acetal sleeve to be used in conjunction with one step-down nose connector and one receptacle (active device).
6. Dimensions apply to sleeve as manufactured. Sleeve material may be compressed when in use.

Inches	mm
.001	0.03
.083	2.11
.084	2.13
.096	2.44
.102	2.59
.108	2.74
.112	2.85
.121	3.07
.122	3.10
.125	3.18
.148	3.76
.152	3.86
.240	6.10
.246	6.25

FIGURE 2. Alignment sleeves.

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Part or Identifying Number (PIN): See table I.

TABLE I. PIN.

PIN M83522/1-	Fiber size µm
03	50/125
03	62.5/125
04	100/140

REQUIREMENTS:

Metals: The plug housing shall be of corrosion resistant steel in accordance with SAE-AMS-QQ-S-763, class 303 or nickel plated brass in accordance with ASTM-B36/B36M alloys C23000, C24000, C26000, or C26800 or ASTM-B16/B16M or ASTM-B122/B122M alloys C377000, C46400, C48200, or C48500. Other metal components shall be of corrosion resistant steel in accordance with SAE-AMS-QQ-S-763, class 303; nickel plated beryllium copper in accordance with ASTM-B196/B196M except alloy C17000, ASTM-B197/B197M, or ASTM-B194 except alloy C17000 or nickel plated brass in accordance with ASTM-B36/B36M alloys C23000, C24000, C26000, or C26800 or ASTM-B16/B16M or ASTM-B122/B122M alloys C377000, C46400, C48200, or C48500.

Epoxies: Use Trabond 230, Eccobond 144B or an equivalent epoxy approved by the qualifying activity.

Dimensions and configuration: See figure 1.

Fiber optic cable requirements:

Cable diameter: In accordance with MIL-PRF-85045.

Cable configuration: In accordance with MIL-PRF-85045.

Fiber diameter: 50/125 µm, 62.5/125 µm, and 100/140 µm.

Fiber numerical aperture: In accordance with MIL-PRF-49291.

Fiber type: Multimode.

Fiber optic contact:

Method of optical alignment: Ferrule/sleeve type (see figure 2).

Lens configuration: Not applicable.

Coating requirements: Not applicable.

Optical requirements:

Number of optical termini: One.

Coupling loss (attenuation): 3 dB maximum per mated pair.

Weight: 6 grams maximum.

Polarization: Not applicable.

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Safety wire holes: Optional.

Force to engage and disengage:

Longitudinal force: 12 pounds maximum.

Torque: 4 inch-pounds maximum.

Coupling proof torque: 5 inch-pounds minimum.

Coupling mechanism retention force: 60 pounds minimum.

Ozone exposure: Not applicable.

Durability: 200 cycles, 1 dB maximum change.

Low temperature: 1 dB change.

Plug accessories: Each plug shall be supplied with a minimum of one crimp ferrule as needed (see figure 1); two alignment sleeves (see figure 2); one protective cable-end cap; one 1.5 inch length of heat-shrink tubing or one rubber boot; and one set of assembly instructions. The assembly instructions shall include a listing of the epoxies specified in this specification sheet and a cleaning procedure statement as follows:

Cleaning procedures: Dampen lens tissue (paper wipes or cotton swab) with a small amount of isopropyl alcohol or water. Gently wipe the face of the connector, removing any debris, particularly around the optical fiber, using clean lens tissue (paper wipes, cotton swab). Dry face of connector by blowing with dry air.

NOTE: The plug mates with MIL-DTL-83522/4 receptacle; MIL-DTL-83522/5 receptacle, MIL-DTL-83522/7 receptacle; MIL-DTL-83522/8 receptacle; or a MIL-DTL-83522/3 adapter mated with another MIL-DTL-83522/1 plug. Connection to any other type plug may result in excessive coupling loss.

Changes from previous issue. The margins of this standard are marked with vertical lines to indicate modifications generated by this change. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-DTL-83522, this document references the following:

MIL-PRF-49291	MIL-DTL-83522/7	ASTM-B36/B36M	ASTM-B197/B197M
MIL-DTL-83522/3	MIL-DTL-83522/8	ASTM-B122/B122M	SAE-AMS-QQ-S-763
MIL-DTL-83522/4	MIL-PRF-85045	ASTM-B194	
MIL-DTL-83522/5	ASTM-B16/B16M	ASTM-B196/B196M	

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